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Propellant-Optimal Powered Descent Guidance Revisited

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This paper presents new and significant advances in propellant-optimal powered descent guidance based on the indirect method. The paper starts by reviewing and improving a recent indirect method relying on analytical equations, and pointing out certain numerical perils of the approach. By taking a fully numerical approach, the developments in this paper take relatively simple and concise software to implement, producing a fully automated algorithm capable of rapidly and reliably generating the complete optimal powered descent trajectory subject to imposed constraints including on the thrust pointing direction, without the need for any empirical assumptions or hard-to-make user-supplied guesses. Furthermore, a "solution degradation" phenomenon not recognized before is brought to attention in the closed-loop solution of the propellant-optimal powered descent problem with a fully constrained final state that can pose a potential safety risk. This risk is mitigated in the proposed guidance approach where guidance in the later half of the powered descent tracks an onboard generated optimal reference trajectory with a globally convergent tracking guidance law.

I. Introduction

Propellant-optimal powered descent guidance provides the guidance commands on thrust magnitude, direction, and flight time to meet the specified touchdown condition with the least amount of propellant usage for the given current condition. The working principles of various methods are invariably reliant on solving the minimum-fuel optimal control problem (in this paper the words "propellant" and "fuel" are used interchangeably). The difference lies in how the optimal control problem is formulated (the extent of detail) and solved numerically. In recent years the representative direct-method based approaches have been those utilizing convex optimization [1, 2] and successive convex optimization [3]. In these approaches the problem is solved as one or a sequence of convex optimization problems. The indirect-method based approaches are exemplified by Ref. [4] where the necessary conditions of the optimal control problem are met by solving a system of nonlinear algebraic system. These two classes of approaches have respective advantages. The direct-method based guidance algorithms are capable of handling various inequality operational constraints and even 6-degree-of-freedom dynamics [3], at higher computational costs and software complexity [5]. The indirect-method based approaches, when solved by a shooting technique, are very efficient computationally and easy to implement with a relatively simple flight software. But adding state inequality constraints and still maintaining convergence robustness can be challenging.

The goal of this paper is to significantly advance the indirect-method based fuel-optimal powered descent guidance approach well beyond Ref. [4]. The indirect approach in Ref. [4] benefits from the theoretically established conclusion that in general in a constant gravity field the fuel-optimal solution consists of no more than 3 bang-off-bang arcs at the upper and lower bounds of the thrust magnitude [4]. While simple, the method in Ref. [4] has a major weakness that the first switching time is determined empirically, and the second switching time found by an outer-loop of univariant optimization. These compromises stem from the desire to avoid the use of the switching function to determine these times because of the perceived numerical complexity with the need for the mass costate in the switching function. Since then, a most recent work in Ref. [6] presents a novel development that offers a simple yet accurate approximation to the switching function for the powered descent problem. In this approximation, the mass costate need not be computed. In its replacement a constant parameter is used in an equivalent expression for the switching function. When this parameter is set to the value of the final mass in the optimal solution, the optimal switching times are found as the roots of a quadratic function of time. While this new finding offers an easy way to determine the switching times, the user is still left to have to guess closely what the optimal final mass is (which is unknown *a priori*) in order to use this new result. As an improvement to the work in Ref. [6], we will automate this selection as an integral part of the solution to the problem so that the user never needs to guess the unknown parameter. With this enhancement, the algorithm proposed in Ref. [6]

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now is in principle capable of finding automatically all the necessary information on the fuel-optimal solution, including the two switching times, total flight time, and thrust direction.

However, we will proceed to point to certain numerical issues that are critical for guidance applications but have been overlooked in Ref. [6]. The solution process in Ref. [6] relies on the piecewise analytical solutions to the equations of motion based on the constant-thrust solution derived in Ref. [7]. The precise values of the times when the piecewise constant thrust switches between the upper and lower bounds are needed. The gradients of these switching times have discontinuity due to the way they are calculated. In another instance the gradients of the switching times do not even exist. When the numerical solution of the problem is sought by using a gradient-based method, the discontinuity and singularity in these gradients can make the convergence more difficult, and frequently result in non-convergence, especially when the initial guess is far from the solution. The singularity of the gradients is an inherent issue of the approach based on analytical solutions of motion that cannot be easily remedied. These pitfalls may not be as critical if the approach is just used to generate optimal powered descent solutions offline. For guidance applications, they pose serious risks if not mitigated.

Taking a philosophically different approach from the existing literature in Refs. [4] and [6] where a preference is placed on closed-form equations, we contend in this work that the aforementioned potential numerical difficulty can be circumvented altogether if we are willing to give up the analytical solutions of the motion in favor of completely numerical solution. In this approach a transformation using a sigmoid function is employed to smooth the bang-bang thrust profile based on the value of the switching function, similar to what is done in Ref. [8]. The switching function still determines the bang-bang optimal thrust structure, only now that the thrust profile switches are no longer instantaneous but continuous (and rapid). No explicit determination of the switching times is necessary anymore. As a result, the problem has no discontinuity or singularity of the gradients anywhere, and a gradient-based method works very well. The propellant penalty caused by the smoothed thrust profile proves to be negligible, and the smoothed thrust profile is closer to being flyable by a real engine. However with the smoothed thrust the problem has no closed-form solutions to the equations of motion, and numerical integration must be used for function evaluations and Jacobian calculations. But once liberated from the limitations of analytical solutions, we realize that the conventional wisdom to avoid computation of the mass costate is unjustified, for the mass costate is numerically well behaved in our problem. With the mass costate available, we can compute the true switching function for determination of the optimal thrust magnitude, without having to rely on an analytically solvable approximation as in Ref. [6] which may not be available in other settings. This freedom enables our algorithm to be more capable as described in the following.

A major algorithmic accomplishment in this paper over Refs. [4] and [6] is an elegant and effective approach to enforce the so-called “pointing constraint" [9] in the indirect method. This constraint limits the direction of the thrust vector, hence the body attitude of the lander. Enforcing such a constraint in the indirect method can significantly complicate the algorithm if not done carefully. In this paper by conducting an analysis of the optimality conditions related to the pointing constraint and leveraging the flexibility of the fully numerical solution approach, we are able to establish a unifying treatment of the problem so that no complexity of the algorithm will incur even if an active pointing constraint is imposed. The implementation is fully automated – the user need not provide any guesses on whether the pointing constraint is expected to be active, where, and how many times. The corresponding optimal thrust magnitude profile (which in general will be different when compared to the case without an active pointing constraint) is also automatically determined by the algorithm.

An important observation in this paper is on a phenomenon we call “solution degradation" that has not been reported in the literature in the context of the fuel-optimal powered descent. When the full trajectory state is constrained at the touchdown (in both position and velocity coordinates), in solving the problem in a closed-loop fashion as would be in guidance applications, trajectory dispersion in the later part of the powered descent can and often does make the solution start to exhibit behaviors drastically different and undesirable from what is expected of a normal powered descent trajectory. This phenomenon arises as a consequence of the highly constrained nature of the problem (fully constrained final state and a constant and maximum thrust toward the end), and is independent of what algorithm is used to solve the problem. Such a trajectory is likely to pose a safety risk, even though it satisfies all the imposed constraints and necessary conditions for optimality. The discovery of this phenomenon suggests that it is not advisable to deploy an algorithm (indirect or direct approaches) to continuously solve a fuel-optimal pinpoint-landing problem all the way to the end for guidance purposes. Instead, the algorithm is used to guide the vehicle from the powered descent initiation (PDI) to an appropriate time before touchdown. Afterwards, the guidance should switch to tracking the last converged optimal solution until touchdown (or the start of the vertical descent phase, if there is one).

For the implementation of this guidance strategy, a simple tracking guidance law is developed by using a predictive control method in Ref. [10]. This tracking law is shown to be able to provide globally asymptotically stable tracking of

the full state (position and velocity) to follow any feasible time-varying trajectory, including the optimal one. This guidance strategy is extensively tested. It proves to be effective in reliably providing high landing precision and optimal propellant performance yet free of the risk of solution degradation.

With all the aforementioned new developments, the algorithm in this paper represents a different and significantly more capable one than what is previously presented in Ref. [4]. We shall refer to this algorithm as "Guidance for Propellant-Optimal Landing of A Rocket" (G-POLAR).

II. Problem Formulation

A. Equations of Motion

In a topocentric Cartesian coordinate frame at the landing site, the three-dimensional equations of motion during powered descent over a flat planet are

$$\begin{aligned} \dot{\boldsymbol{r}} &= \boldsymbol{V}, & \boldsymbol{r}(t_0) &= \boldsymbol{r}_0 \\ \dot{\boldsymbol{V}} &= \frac{T_{thrust}}{m}\mathbf{1}_T + \boldsymbol{g}, & \boldsymbol{V}(t_0) &= \boldsymbol{V}_0 \\ \dot{m} &= -\frac{T}{V_e}, & m(t_0) &= m_0 \end{aligned} \tag{1}$$

where $\boldsymbol{r} = [x\ y\ z]^T$ is the position vector, $\boldsymbol{V} = [u\ v\ w]^T$ the velocity vector, and $\boldsymbol{g} = [g_x\ g_y\ g_z]^T$ is the gravitational acceleration vector which is taken to be a constant with a magnitude of g_0. The rocket engine thrust magnitude is T_{thrust} and it is constrained by given upper and lower bounds

$$\hat{T}_{min} \leq T_{thrust} \leq \hat{T}_{max} \tag{2}$$

where $\hat{T}_{min} > 0$, as the engine should not be completely shut down in the relatively short duration of powered descent. The exhaust velocity of the engine is $V_e > 0$ and will be treated as a constant in this work, and $\mathbf{1_T}$ is a unit vector that defines the thrust direction. The controls in this problem are $\mathbf{1}_T$ and T_{thrust}. The consideration of the pointing constraint is deferred to Section VIII later.

Appropriate scaling of variables is critical to aid reliable convergence of the algorithm. For this purpose, we will choose three scaling factors R_{scale}, t_{scale}, and m_{scale} for lengths, time, and mass, respectively

$$\begin{aligned} R_{scale} &= R_0 > 0\ (m) \\ t_{scale} &= \sqrt{\frac{R_{scale}}{g_0}}\ (s) \\ V_{scale} = R_{scale}/t_{scale} &= \sqrt{g_0 R_{scale}} \\ m_{scale} &= m_0\ (kg) \end{aligned} \tag{3}$$

where g_0 is the gravitational acceleration at the equator of the planet. The choice of the value of $R_{scale} = R_0$ may be used as a "knob" to achieve better balance of the orders of magnitudes of the entries of the Jacobian of the problem for better convergence. An appropriate choice may depend on the coordinate system used and destination planet. For instance when a topocentric coordinate frame at the landing site is used for powered descent at Mars, the position coordinates are typically on the order of several kilometers at the start of powered descent. Therefore a reasonable value for R_{scale} may be

$$R_{scale} = 10,000 \quad (m) \tag{4}$$

Let us normalize all the dimensional variables and parameters in Eq. (1) by the respective scaling factors defined above (where (V_e is normalized by V_{scale} and $\boldsymbol{g}$ by g_0), and define the normalized (dimensionless) thrust acceleration (in g_0) by

$$T = \frac{T_{thrust}}{m_{scale} g_0} \tag{5}$$

With some abuse of notation where we continue to use the same symbols for the nondimensional variables and

Journal of Guidance, Control, and Dynamics, Vol. 46, No. 2, pp. 215–230, 2023; https://doi.org/10.2514/1.G007214

parameters, the nondimensional equations of motion are

$$\begin{aligned} \dot{\boldsymbol{r}} &= \boldsymbol{V}, & \boldsymbol{r}(t_0) &= \boldsymbol{r}_0 \\ \dot{\boldsymbol{V}} &= \frac{T}{m}\mathbf{1}_T + \boldsymbol{g}, & \boldsymbol{V}(t_0) &= \boldsymbol{V}_0 \\ \dot{m} &= -\frac{T}{V_e}, & m(t_0) &= 1.0 \end{aligned} \tag{6}$$

The constraints on the controls $\{T, \mathbf{1}_T\}$ are

$$\mathbf{1}_T^T \mathbf{1}_T = 1 \tag{7}$$

$$T_{min} \leq T \leq T_{max} \tag{8}$$

where

$$T_{min} = \hat{T}_{min}/m_{scale}g_0, \quad T_{max} = \hat{T}_{max}/m_{scale}g_0 \tag{9}$$

B. Optimal Powered Descent Problems

Following Ref. [4], we consider 3 variations of the propellant-optimal powered descent problem:

Pinpoint-Landing Problem

The performance index to minimize is the propellant consumption. For a given initial mass, it is equivalent to maximization of the final mass

$$J = -m(t_f) \tag{10}$$

where the final time $t_f > t_0$ is free. The terminal state is fully constrained by

$$\begin{aligned} \boldsymbol{r}(t_f) &= \boldsymbol{r}_f^* \\ \boldsymbol{V}(t_f) &= \boldsymbol{V}_f^* \end{aligned} \tag{11}$$

where $\boldsymbol{r}_f^*$ and $\boldsymbol{V}_f^*$ are specified and represent the desired final condition of the powered descent phase. In a topocentric frame with the origin at the landing site, $\boldsymbol{r}^* = 0$ implies landing at the designated site.

Close-Approach Landing Problem

When it is not possible to land on the designated landing site due to severely off-nominal condition, or in a scenario where landing close to the specified location is sufficient, the first guidance priority is to land on the ground softly, and the second priority is to land sufficiently close to the designated site, while using the least amount of propellant doing it. We call this the close-approach landing problem (in Ref. [4] it is called the Bolza landing problem). For specificity of presentation we will assume that a North-Up-East (NUE) topocentric frame is used where the y-axis is in the local vertical (up) direction. Figure 1 depicts this frame and its relationship with a planet-center planet-fixed frame $OXYZ$ which has its Z-axis point to the North Pole. In lieu of the complete pinpoint-landing conditions in Eq. (11), the terminal constraints for the close-approach landing problem in the NUE frame are now

$$y(t_f) = 0 \tag{12}$$

$$\boldsymbol{V}(t_f) - \boldsymbol{V}_f^* = 0 \tag{13}$$

The performance index is now modified to be

$$J = \frac{\kappa}{2}\left[(x(t_f) - x_f^*)^2 + (z(t_f) - z_f^*)^2\right] - m(t_f) \tag{14}$$

where x_f^* and z_f^* are given components of $\boldsymbol{r}_f^*$ (for instance $x_f^* = z_f^* = 0$); $\kappa \geq 0$ is a weighting coefficient that penalizes the landing distance to the location (x_f^*, z_f^*) on the ground.

In Ref. [2] a minimum-landing-error problem is formulated where the performance index penalizes only the landing error but without the propellant consumption term as in Eq. (14) (hence it is called “minimum-error" problem and the current problem only claims "close-approach" solution). The difference between the two problems is not trivial.

Soft-Landing Problem

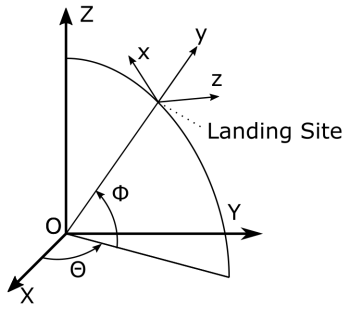


Fig. 1 North-Up-East (NUE) topocentric frame xyz at the landing site

When the touchdown location is not of any concern, but the priority is to land softly on the ground, at any location, with the minimum propellant consumption, this is called the soft-landing problem. The soft-landing problem is a special case of the close-approach landing problem when $\kappa = 0$ in the performance index in Eq. (14). On the other hand, when κ in the close-approach landing problem is very large, the solution may be practically regarded to be a close approximation to the solution of the pinpoint-landing problem. The reader is referred to Ref. [4] for a theoretical exposition on the comparison of propellant consumption in these 3 problems.

The convergence of the solution process for the soft-landing and close-approach landing problems with a relatively small to medium κ is very robust because there are no hard constraints on the landing location. Soft-landing/close-approach landing solutions have been used to effectively predict a feasible and fuel-efficient PDI condition [4, 11].

III. Necessary Conditions

A. Optimal Control

By the Maximum Principle [12], the necessary conditions that must be satisfied by the optimal powered descent problem are as follows, facilitated by the Hamiltonian

$$H = \boldsymbol{p}_r^T \boldsymbol{V} + \boldsymbol{p}_V^T \left(\frac{T}{m} \mathbf{1}_T + \boldsymbol{g} \right) - p_m \frac{T}{V_e} = \boldsymbol{p}_r^T \boldsymbol{V} + \boldsymbol{p}_V^T \boldsymbol{g} + T \left(\frac{\boldsymbol{p}_V^T \mathbf{1}_T}{m} - \frac{p_m}{V_e} \right) \tag{15}$$

where $\boldsymbol{p}_r \in R^3$ and $\boldsymbol{p}_V \in R^3$ are the position and velocity costate vectors, and p_m the mass costate. The position and velocity costate vectors satisfy

$$\begin{aligned} \dot{\boldsymbol{p}}_r &= -\frac{\partial H}{\partial \boldsymbol{r}} = 0 \qquad \rightarrow \qquad \boldsymbol{p}_r = \boldsymbol{\lambda} \\ \dot{\boldsymbol{p}}_V &= -\frac{\partial H}{\partial \boldsymbol{V}} = -\boldsymbol{p}_r \qquad \rightarrow \qquad \boldsymbol{p}_V = -\boldsymbol{\lambda}(t - t_0) + \boldsymbol{c} \end{aligned} \tag{16}$$

where $\boldsymbol{\lambda} = [\lambda_1 \ \lambda_2 \ \lambda_3]^T$ and $\boldsymbol{c} = [c_1 \ c_2 \ c_3]^T$ which are both constant vectors. The optimal unit vector $\mathbf{1}_T$ that maximizes the Hamiltonian is given by

$$\mathbf{1}_T^* = \arg \max_{\mathbf{1}_T, \|\mathbf{1}_T\|=1} H(\boldsymbol{V}, m, \boldsymbol{p}_r, \boldsymbol{p}_V, T, \mathbf{1}_T) = \frac{\boldsymbol{p}_V}{\|\boldsymbol{p}_V\|} = \frac{-\boldsymbol{\lambda}(t - t_0) + \boldsymbol{c}}{\sqrt{p(t - t_0)^2 + q(t - t_0) + r}} \tag{17}$$

where, using the same notation as in Ref. [7], we have

$$\begin{aligned} p &= \boldsymbol{\lambda}^T \boldsymbol{\lambda} = \lambda_1^2 + \lambda_2^2 + \lambda_3^2 \\ q &= -2\boldsymbol{\lambda}^T \boldsymbol{c} = -2(\lambda_1 c_1 + \lambda_2 c_2 + \lambda_3 c_3) \\ r &= \boldsymbol{c}^T \boldsymbol{c} = c_1^2 + c_2^2 + c_3^2 \end{aligned} \tag{18}$$

Define the switching function S to be the quantity in the parentheses after T in the Hamiltonian in Eq. (15)

$$S = \frac{\boldsymbol{p}_V^T \mathbf{1}_T}{m} - \frac{p_m}{V_e} \tag{19}$$

Using the optimal thrust direction $\mathbf{1}_T^*$ from Eq. (17), the explicit expression of the switching function is

$$S = \frac{\|\boldsymbol{p}_V\|}{m} - \frac{p_m}{V_e} = \frac{\sqrt{p(t-t_0)^2 + q(t-t_0) + r}}{m} - \frac{p_m}{V_e} \tag{20}$$

The optimal thrust magnitude is bang-bang

$$T^* = \begin{cases} T_{max}, & S > 0 \\ T_{min}, & S < 0 \end{cases} \tag{21}$$

Reference [4] establishes that in a constant-gravity field, there is no physically possible case in powered descent where $S \equiv 0$ in any finite period of time (i.e., no possibility of singular optimal control), and the optimal thrust magnitude is in general composed of three thrust arcs with piecewise constant profile of T_{max}-T_{min}-T_{max} as shown in Fig. 2.

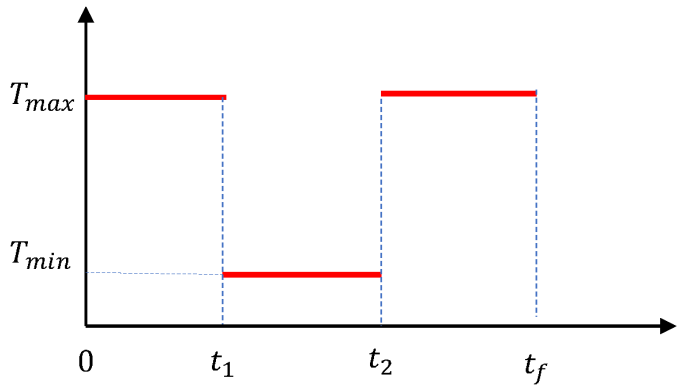


Fig. 2 Fuel-optimal thrust magnitude profile (cf. Ref [4]).

B. Approximate Switching Function

The mass costate p_m is governed by the differential equation

$$\dot{p}_m = -\frac{\partial H}{\partial m} = \frac{T^* \boldsymbol{p}_V^T \mathbf{1}_T^*}{m^2} = \frac{T^* \sqrt{p(t-t_0)^2 + q(t-t_0) + r}}{m^2} \tag{22}$$

This equation suggests that $p_m(t)$ is monotonically increasing. But there is no known closed-form solution for $p_m(t)$ that can be used in the switching function S in Eq. (20). Most recently it is shown in Ref. [6] that for a fuel-optimal powered descent problem that satisfies the following conditions:

1) $V(t_f) \approx 0$;
2) The final thrust acceleration is greater or equal to $2g_0$;
3) The mass of the propellant consumed is sufficiently smaller than the final mass of the vehicle;

the product $m(t)p_m(t)$ is approximately constant for all $t \in [t_0, t_f]$. Using the transversality condition that $p_m(t_f) = 1$, this means that $m(t)p_m(t) \approx m(t_f)$. Denote

$$K_c = m(t_f) \tag{23}$$

Consider the following function

$$\hat{K}(t) = p(t-t_0)^2 + q(t-t_0) + r - \left(\frac{K_c}{V_e}\right)^2 \tag{24}$$

Compare $\hat{K}$ with the switching function S in Eq. (20), if we take $m(t)p_m(t) = m(t_f) = K_c$, $\hat{K} = 0$ and $S = 0$ have the identical roots. Furthermore, $\hat{K}$ and S have the same sign at the same t. Therefore the optimal bang-bang thrust can be

Journal of Guidance, Control, and Dynamics, Vol. 46, No. 2, pp. 215–230, 2023; https://doi.org/10.2514/1.G007214

determined by $\hat{K}$ instead

$$T^* = \begin{cases} T_{max}, & \hat{K} > 0 \\ T_{min}, & \hat{K} < 0 \end{cases} \tag{25}$$

Since $p > 0$, the quadratic function of time $\hat{K}$ has a unique minimum at $t = -q/(2p)$. The approximate switching function in Eq. (24) allows analytical calculation and prediction of the possible switching times at any given constant $\boldsymbol{\lambda}$, $\boldsymbol{c}$ and K_c. But since $m(t_f)$ is not known *a priori* and it will be different even for the same mission if the PDI condition differs, Ref. [6] treats K_c as a user-guessed constant. We will describe a way in Section V to relieve this burden on the user and fully automate the correct choice of K_c as a part of the problem solution for any mission and any actual condition.

C. Transversality Conditions

Since the final time t_f is free and the performance index is $J = -m(t_f)$ or Eq. (14), the transversality condition for the Hamiltonian is

$$H(t_f) = \frac{\partial J}{\partial t_f} = 0 \tag{26}$$

Additionally, by the Maximum Principle the transversality condition on p_m is,

$$p_m(t_f) = -\frac{\partial[-m(t_f)]}{\partial m(t_f)} = 1 \tag{27}$$

Combining both Eqs. (15), (26), and (27), the expression for $H(t_f)$ is

$$H(t_f) = \boldsymbol{p}_r^T(t_f)\boldsymbol{V}(t_f) + \boldsymbol{p}_V^T(t_f)\left(\frac{T(t_f)}{m(t_f)}\mathbf{1}_T(t_f) + \boldsymbol{g}\right) - \frac{T(t_f)}{V_e} = 0 \tag{28}$$

Using Eqs. (10), (16), (17) at the final time in Eq. (28) yields,

$$H(t_f) = \boldsymbol{\lambda}^T\boldsymbol{V}_f^* + \frac{T_{max}}{m(t_f)}\sqrt{p(t_f - t_0)^2 + q(t_f - t_0) + r} + (-\boldsymbol{\lambda}(t_f - t_0) + \boldsymbol{c})^T\boldsymbol{g} - \frac{T_{max}}{V_e} = 0 \tag{29}$$

where $T^*(t_f) = T_{max}$ has been used [4].

The pinpoint-landing problem has no other transversality conditions. For the close-approach landing problem, let

$$\phi = \frac{\kappa}{2}\left[(x(t_f) - x_f^*)^2 + (z(t_f) - z_f^*)^2\right]$$

The two transversality conditions independent of any Lagrange multipliers are on $p_{r_x}(t_f)$ and $p_{r_z}(t_f)$

$$\begin{aligned} p_{r_x}(t_f) &= -\frac{\partial\phi}{\partial x_f} \\ p_{r_z}(t_f) &= -\frac{\partial\phi}{\partial z_f} \end{aligned} \tag{30}$$

Using $p_{r_x} = \lambda_1$ and $p_{r_z} = \lambda_3$ and evaluating the partial derivatives of ϕ, we have

$$\begin{aligned} \lambda_1 + \kappa[x(t_f) - x_f^*] &= 0 \\ \lambda_3 + \kappa[z(t_f) - z_f^*] &= 0 \end{aligned} \tag{31}$$

The above transversality conditions become the transversality conditions for the soft-landing problem if $\kappa = 0$. Note that if a different topocentric coordinate frame is used where the vertical axis is not the y-axis, the first condition in Eq. (12), the penalty term in Eq. (14), and one of the equations in Eq. (31) should be changed accordingly.

IV. Solution with Analytical Equations

In this section the solution approach relying on the analytical expressions of all relevant equations are outlined.

A. Determination of Switching Times

For given p, q, r and K_c, let t_1^* and t_2^* be the two roots of the quadratic function of time [6]

$$\hat{K}(t) = p(t-t_0)^2 + q(t-t_0) + r - \left(\frac{K_c}{V_e}\right)^2 = 0 \tag{32}$$

If t_1^* and t_2^* are complex conjugates, $\hat{K}$ does not change sign, and there is no switching of the thrust magnitude. If t_1^* and t_2^* are real, denote $t_1^* \le t_2^*$. Since $\hat{K}(t)$ is a convex function of t, $\hat{K}(t) < 0$ in the interval (t_1^*, t_2^*). Because the true switching times t_1 and t_2 must be in the interval $[t_0, t_f]$ for the solution to be meaningful, the standard practice is to set t_1 to be equal to t_1^* only when $t_1^* \in [t_0, t_f]$, and $t_2 = t_2^*$ only when $t_2^* \in [t_0, t_f]$. When either t_1^* and/or t_2^* are outside $[t_0, t_f]$, t_1 and t_2 are truncated to be t_0 or t_f, depending on where t_1^* and/or t_2^* lie. See Ref. [6] for detail. Once t_1 and t_2 are found, we know that $T = T_{max}$ in $[t_0, t_1)$, $T = T_{min}$ in $[t_1, t_2)$, and $T = T_{max}$ in $[t_2, t_f]$.

As $\boldsymbol{\lambda}$, $\boldsymbol{c}$, and K_c are being updated, t_1^* and t_2^* will change, as well as t_1 and t_2. If during the iterations t_1^* and t_2^* are outside or move out of the interval $[t_0, t_f]$, the actual switching times t_1 and t_2 are kept at t_0 or t_f because setting them equal to t_1^* and t_2^* would violate the order of $0 \le t_1 \le t_2 \le t_f$ (cf. Fig. 2). When t_1 and t_2 are truncated at t_0 or t_f, they no longer change with $\boldsymbol{\lambda}$, $\boldsymbol{c}$, and K_c as t_1^* and t_2^* still do. This implies that the gradients of t_1 and t_2 with respect to $\boldsymbol{\lambda}$, $\boldsymbol{c}$, and K_c are zero as soon as t_1 and t_2 are at t_0 or t_f, leading to discontinuities in the gradients of t_1 and t_2. It is stressed that these discontinuities are present, no matter whether the gradients are calculated analytically or numerically. These discontinuities can cause convergence difficulty for a gradient-based search method to update $\boldsymbol{\lambda}$, $\boldsymbol{c}$, and K_c. This effect is discussed more and remedies are proposed in Section VI.

B. Analytical Solutions with Piecewise Constant Thrust

In each of the 3 possible arcs of the optimal solution, the optimal thrust T^* is a constant (Fig. 2). With the thrust direction given by Eq. (17), closed-form solution to the equations of motion in Eq. (6) in each arc can be obtained [7] for given $\boldsymbol{\lambda}$, $\boldsymbol{c}$, t_1, t_2, t_f, engine parameter V_e, and initial conditions. The reader is referred to Ref. [7] for specifics on the solution over a constant-thrust arc. Applying the equations for the solution over the 3 arcs with piecewise constant thrust magnitude. the final state at t_f is given below (with $t_0 = 0$) in a compact form

$$\begin{aligned}
\boldsymbol{V}(t_f) =& \boldsymbol{V}_0 + V_e\left(\sum_{i=1}^{3} B(t_i, t_{i-1})\right)\boldsymbol{\lambda} - V_e\left(\sum_{i=1}^{3} A(t_i, t_{i-1})\right)\boldsymbol{c} + t_f\boldsymbol{g} \\
\boldsymbol{r}(t_f) =& \boldsymbol{r}_0 + t_f\boldsymbol{V}(t_f) - V_e\left(\sum_{i=1}^{3} C(t_i, t_{i-1})\right)\boldsymbol{\lambda} + V_e\left(\sum_{i=1}^{3} B(t_i, t_{i-1})\right)\boldsymbol{c} - \frac{t_f^2}{2}\boldsymbol{g} \\
m(t_f) =& m_0 - \frac{(T_{max} - T_{min})}{V_e}t_1 + \frac{(T_{max} - T_{min})}{V_e}t_2 - \frac{T_{max}}{V_e}t_f
\end{aligned} \tag{33}$$

where $t_3 = t_f$ (and the "+" sign in front of the t_2 term in the equation for $m(t_f)$ is not a typo). The analytical expressions of the thrust integral terms $A(t_i, t_{i-1})$, $B(t_i, t_{i-1})$, and $C(t_i, t_{i-1})$ are not only functions of $t_i, i = 1, 2, 3$, but also functions of $\boldsymbol{\lambda}$ and $\boldsymbol{c}$. The detailed expressions and equations can be found in Ref. [13].

V. The Zero-Finding Problems

To completely determine the solution of the problem, we need not only know $\boldsymbol{\lambda}$, $\boldsymbol{c}$, and t_f, but K_c as well, since it helps define the switching times t_1 and t_2. We will treat K_c as the 8th unknown, in addition to $\boldsymbol{\lambda}$, $\boldsymbol{c}$, and t_f. For the pinpoint-landing problem, the 8 equality constraints required to be satisfied by the unknowns $(\boldsymbol{\lambda},\ \boldsymbol{c},\ t_f,\ K_c)$ are

$$\boldsymbol{r}(t_f) - \boldsymbol{r}_f^* = 0 \tag{34}$$

$$\boldsymbol{V}(t_f) - \boldsymbol{V}_f^* = 0 \tag{35}$$

$$\boldsymbol{\lambda}^T\boldsymbol{V}_f^* + \frac{T}{m(t_f)}\sqrt{pt_f^2 + qt_f + r} + (-\boldsymbol{\lambda}t_f + \boldsymbol{c})^T\boldsymbol{g} - \frac{T_{max}}{V_e} = 0 \tag{36}$$

$$K_c - m(t_f) = 0 \tag{37}$$

Equation (37) comes from the requirement in Eq. (23). Now instead of a user-supplied guess as required in Ref. [6], we find the correct value of K_c as a part of the solution to the problem by the algorithm. Constraint (36) is from

the transversality condition in Eq. (28). Given the closed-form solutions in Eq. (33), every constraint in the system Eqs. (34)–(37) has a closed-form expression.

For the close-approach landing problem, the 8 constraints are

$$y(t_f) = 0 \tag{38}$$

$$\lambda_1 + \kappa[x(t_f) - x_f^*] = 0 \tag{39}$$

$$\lambda_3 + \kappa[z(t_f) - z_f^*] = 0 \tag{40}$$

$$\boldsymbol{V}(t_f) - \boldsymbol{V}_f^* = 0 \tag{41}$$

$$\boldsymbol{\lambda}^T \boldsymbol{V}_f^* + \frac{T}{m(t_f)}\sqrt{pt_f^2 + qt_f + r} + (-\boldsymbol{\lambda} t_f + \boldsymbol{c})^T \boldsymbol{g} - \frac{T_{max}}{V_e} = 0 \tag{42}$$

$$K_c - m(t_f) = 0 \tag{43}$$

When $\kappa = 0$, Eqs. (38)–(43) are also the constraints for the soft-landing problem.

Now the problem becomes a zero-finding problem for a system of 8 nonlinear algebraic equations of the vector $z = (\boldsymbol{\lambda}^T,\ \boldsymbol{c}^T,\ t_f,\ K_c) \in R^8$, abstracted to be

$$\boldsymbol{f}(z) = 0 \tag{44}$$

The Jacobian of the system in Eq. (44) can also be obtained analytically [13] (albeit a tedious, but straightforward undertaking) if a gradient-based method is to be used to solve the problem.

To start the numerical solution process, an initial guess to $\boldsymbol{z}$ is usually needed. Since t_f and K_c have physical meanings, it is easy to make a guess. The vector $\boldsymbol{c}$ is the initial value of the primer vector $\boldsymbol{p}_V(t)$, and it defines the initial thrust direction. The following initial guess is reasonable

$$\boldsymbol{c} = -\beta \frac{\boldsymbol{V}_0}{\|\boldsymbol{V}_0\|}, \quad \beta > 0 \tag{45}$$

where β is a scaling coefficient that may be chosen to bring the order of magnitude of the initial guess for $\boldsymbol{c}$ closer to the converged one.

A reasonable choice of the initial guess on $\boldsymbol{\lambda}$ may be made based on the particular coordinate system used. Consider the NUE topocentric coordinate system in Fig. 1. In the chosen coordinate system, one can expect that the final thrust direction, thus the direction of $\boldsymbol{p}_V(t_f)$, is pointing upward, that is, in the positive y-direction. Thus, for $\boldsymbol{p}_V(t_f) = [p_{V_1}(t_f) \quad p_{V_2}((t_f) \quad p_{V_3}((t_f)]^T$, we expect approximately

$$\begin{aligned} p_{V_1}(t_f) &= -\lambda_1 t_f + c_1 \approx 0 \\ p_{V_3}(t_f) &= -\lambda_3 t_f + c_3 \approx 0 \end{aligned} \tag{46}$$

Then, two reasonable initial guesses are

$$\lambda_1 = \frac{c_1}{t_f}, \quad \lambda_3 = \frac{c_3}{t_f} \tag{47}$$

where c_1 and c_3 are already defined in Eq. (45). Because the final thrust is expected to have a positive vertical component, we should have $p_{V_2}(t_f) = -\lambda_2 t_f + c_2 > 0$. Choose $\lambda_2 = \eta \frac{c_2}{t_f}$, where η is a scaling factor chosen according to the following rules

$$\begin{cases} \eta < 1; & \text{if } c_2 \geq 0 \\ \eta > 1; & \text{if } c_2 < 0 \end{cases} \tag{48}$$

The above choice of η ensures that $p_{V_2}(t_f) = (1 - \eta)c_2 > 0$.

It should be stressed that the first 7 equations in the zero-finding problems in Eqs. (34)–(37) and in Eqs. (38)–(43) remain unchanged even if the method of solution does not rely on the analytical solutions in Section IV and the approximate switching function $\hat{K}$ is not used. In Section VIII a fully numerical but much more flexible and robust solution approach will be presented that will still solve the above zero-finding problems with conditions (37) and (43) replaced by condition (27).

VI. Where Things May Fall Apart

The numerical solution approach based on Sections IV.A and IV.B benefits from analytical equations on all needed conditions, and at first glance seems a rare lucky break for a formulation of reasonable fidelity for a practical problem. This is the solution approach used in Ref. [6]. But upon a closer examination one is confronted with the possible discontinuities in $\partial t_1/\partial z$ and $\partial t_2/\partial z$ identified in Section IV.A. The other situation where the gradients $\partial t_1/\partial z$ and $\partial t_2/\partial z$ do not even exist is discussed in the following.

The roots of Eq. (32) are given by

$$t^*_{1,2} = \frac{-q \pm \sqrt{q^2 - 4p[r - (K_c/Ve)^2]}}{2} \tag{49}$$

At the point where

$$D_{sw} = q^2 - 4p[r - (K_c/Ve)^2] = 0 \tag{50}$$

$\partial t^*_1/\partial z$ and $\partial t^*_2/\partial z$ are undefined, hence neither $\partial t_1/\partial z$ and $\partial t_2/\partial z$. This situation will potentially occur for instance when the initial guess is such that $D_{sw} < 0$, but the optimal solution calls for a $t_1 > 0$ and $t_2 > 0$. In such a case D_{sw} will have to cross the point where $D_{sw} = 0$ into the region where $D_{sw} > 0$. As a result a gradient-based method will likely fail. If this is an algorithm to be used for guidance purposes, one cannot defend this singularity by arguing that the chance for an iteration to be right at $D_{sw} = 0$ is small. In fact we have observed that whenever the solution process is in a situation where D_{sw} has to change sign to reach the solution, failure to converge is likely.

The discontinuity and possible singularity in the gradients of the switching times will cause the Jacobian of the system in Eq. (44), $\partial \boldsymbol{f}/\partial z$, to be discontinuous or undefined. The consequence is a slow convergence of a gradient-based method for solving the root-finding problem in Eq. (44), or failure to converge. For instance, when a Newton-Raphson method with inexact line-search is used to solve a particular pinpoint-landing problem, Fig. 3 shows the values of t_1, t_2 and t_f during the iterations, starting from an "uneducated" initial guess. The initial guess produces an initial solution where t_1 and t_2 are both truncated at $t_0 = 0$ where the discontinuities in the Jacobian occur. It takes 13 iterations for t_1 and t_2 to move away from $t_0 = 0$, and another 25 iterations (for a total of 38 iterations) to finally converge to their respective non-zero values. The struggle to find the solution is palpable.

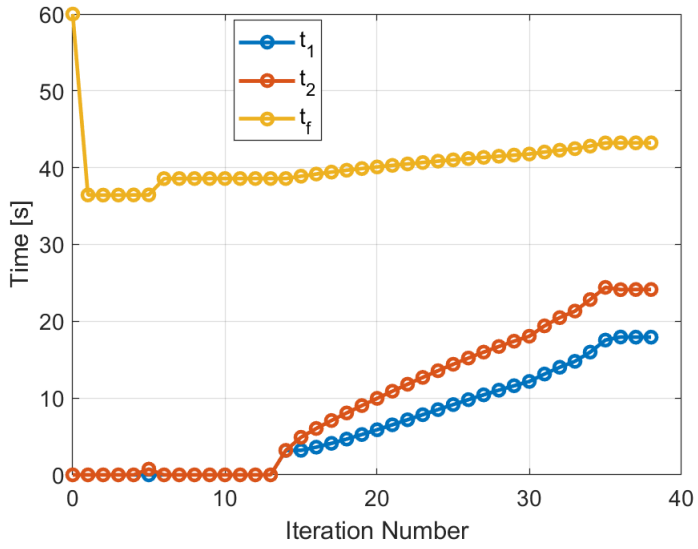


Fig. 3 Iterates in t_1, t_2 and t_f from a starting point where the discontinuities in the Jacobian cause slow convergence

In the following, a technique to eliminate the discontinuity in the gradients of t_1 and t_2 due to the non-smooth truncation is described. The idea is to find a smooth mapping that maps t^*_1 and t^*_2, the two real roots of Eq. (32), into the close interval $[0, t_f]$ to define the corresponding t_1 and t_2. While t^*_1 and t^*_2 can be anywhere, the switching times t_1 and t_2 must be strictly in the closed interval $[0, t_f]$ – any excursion outside this interval would render the solution

meaningless. For this reason, we will use the following technique that will eliminate the said gradient discontinuity of t_1 and t_2 and ensure t_1 and t_2 to strictly in $[0, t_f]$. The idea is to design a saturation function that uses cubic splines to make smooth transition from the linear into the saturation regions. Define

$$\delta > 0,\ x_1 = \delta,\ x_2 = 1 - \delta \tag{51}$$

where $\delta \in (0, 1)$ is a small number (e. g., $0.03 \sim 0.05$) that defines the thickness of the "boundary transition layers" at the ends of the linear region of the saturation function - see below. Furthermore, denote

$$a_2 = 2/x_1 \tag{52}$$

$$a_3 = -1/x_1^2 \tag{53}$$

$$b_2 = 1/(1 - x_2) \tag{54}$$

$$b_3 = -1/(1 - x_2)^2 \tag{55}$$

The smooth saturation function is designed to be

$$s(x) = \begin{cases} 0, & x < 0 \\ a_3x^3 + a_2x^2, & 0 \le x \le x_1 \\ x, & x_1 < x \le x_2 \\ b_3(x - x_2)^3 + b_2(x - x_2)^2 + x, & x_2 < x \le 1 \\ 1, & x > 1 \end{cases} \tag{56}$$

Unlike a standard saturation function, it can be readily shown that $s(x)$ has a continuous first-order derivative everywhere, particularly at $x = 0$ and $x = 1$ where a standard saturation function is not differentiable. Also unlike other smooth sigmoid functions, $s(x)$ strictly and accurately realizes the saturation conditions $s(0) = 0$ and $s(1) = 1$. Figure 4 plots this function with $\delta = 0.03$. Notice the smooth transition at $x = 0$ and $x = 1$.

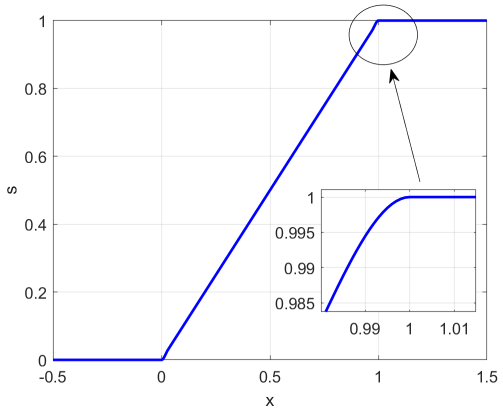


Fig. 4 Smooth saturation function defined in Eq. (56)

Consider the case when Eq. (32) has two real roots t_1^* and t_2^* as given in Eq. (49). The actual switching times t_1 and t_2 will be determined by using the function in Eq. (56) as follows. Let

$$\tau_i = \frac{t_i^*}{t_f}, \quad i = 1, 2 \tag{57}$$

The switching times t_1 and t_2 are then found from

$$t_i = t_f s(\tau_i), \quad i = 1, 2 \tag{58}$$

No other conditional logic is needed. When t_1^* and t_2^* are in $[0, t_f]$, Eq. (58) gives $t_1 = t_1^*$ and $t_2 = t_2^*$; when $t_1^* < 0$ and $t_2^* < 0$, Eq. (58) leads to $t_1 = 0$ and $t_2 = 0$; when $t_1^* > t_f$ and $t_2^* > t_f$, Eq. (58) yields $t_1 = t_f$ and $t_2 = t_f$. Most

importantly, t_1 and t_2 so defined now have continuous gradients with respect the unknown vector z. The gradients are given by

$$\frac{\partial t_i}{\partial z} = s(\tau_i)\frac{\partial t_f}{\partial z} + t_f \frac{ds(\tau_i)}{d\tau_i}\frac{\partial \tau_i}{\partial z}, \quad i = 1, 2 \tag{59}$$

where $z \in R^8$ is the unknown vector, and

$$\frac{\partial \tau_i}{\partial z} = \frac{1}{t_f}\frac{\partial t_i^*}{\partial z} - \frac{t_i^*}{t_f^2}\frac{\partial t_f}{\partial z}, \quad i = 1, 2 \tag{60}$$

The gradients $\partial t_i^*/\partial z$ exist and are continuous everywhere, except at the point where $D_{sw} = 0$ and $\partial t_i^*/\partial z$'s do not exist.

All the pitfalls discussed in this subsection arise from the need to determine and use t_1 and t_2 explicitly because the analytical solutions in Section IV.B require the precise knowledge of t_1 and t_2. If we do not rely on the analytical solutions but numerical solutions instead, all the above described issues can be avoided completely, and a more reliable version of the algorithm may emerge. This appealing prospect motivates the next section.

VII. Solution with Smoothed Thrust

The centerpiece algorithm in G-POLAR departs from the analytically centric approaches where the closed-form solutions of the motion play the role of an anchor. While the working principle remains the same, to solve the root-finding problems defined in Section V, a numerically centric approach is taken in this section to completely eliminate the potential difficulties associated with computing t_1 and t_2 explicitly as discussed in Section VI, including those caused by the discontinuities in the Jacobian and the case where the gradients of t_1 and t_2 do not exist. In this approach every necessary condition in Section III remains in force. But the implementation of the optimal bang-bang thrust solution in Eq. 25 is realized by a smoothed close approximation by using a sigmoid function. Hyperbolic tangent function is a popular sigmoid function, as proposed in Ref. [8], among other possibilities. Following Ref. [8], we model the optimal thrust by

$$T^*(t) = \frac{1}{2}\left(T_{max} + T_{min}\right) + \frac{1}{2}\left(T_{max} - T_{min}\right)\tanh\left[\frac{\hat{K}(t)}{1-\varepsilon}\right], \quad 0 < \varepsilon < 1 \tag{61}$$

where $\hat{K}(t)$ is defined in Eq. (24). The hyperbolic tangent function approaches ± 1 as its argument approaches $\pm\infty$. When the parameter ε is chosen sufficiently close to 1, the thrust magnitude given by Eq. (61) will be practically T_{max} if $\hat{K}(t) > 0$, and T_{min} if $\hat{K}(t) < 0$, consistent with Eq. (25). But there are two major differences induced by this approximation. The first is that the thrust magnitude now is not strictly bang-bang, but varies continuously (and rapidly) from one bound to another when $\hat{K}(t)$ changes sign. The second is more pertinent to our purpose, that is, the switching times t_1 and t_2 no longer need to be determined and explicitly used to define the state solution. The thrust given by Eq. (61) moves automatically to T_{max} or T_{min} according to the sign of $\hat{K}(t)$. As a result, the problem is now continuously differentiable with respect to z, and all potential numerical difficulties associated the switching times t_1 and t_2 discussed in Section VI are circumvented competely. A gradient-based method such as Newton-Raphson works well in solving the root-finding problem.

With this setting, the 8 constraints for the unknown vector z in Eqs. (34)–(37) for the pinpoint-landing problem remain valid, as well as Eqs. (38)–(43) for the close-approach landing and soft-landing problems. But the closed-form solutions for the trajectory state in Section IV.B are no longer applicable. And the equations of motion with the the thrust in Eq. (61) are not analytically integrable. Therefore the evaluations of the function in Eq. (44) are now made by numerically integrating the equations of motion (6) to get $\boldsymbol{r}(t_f)$, $\boldsymbol{V}(t_f)$, and $m(t_f)$, Similarly, the Jacobian of $\boldsymbol{f}(z)$ can only be evaluated numerically by finite difference (which requires numerical integration of the equations of motion). In this approach the computational requirements are higher than that with the analytical solutions and Jacobian, but still well within the capability of a modern flight computer. As a quantitative illustration, when the algorithm is implemented in FORTRAN 90 and executed on a 7-year desktop computer with an Intel Xeon 2.10GHz CPU, the wall clock time measured by FORTRAN intrinsic function *system_clock* for G-POLAR to find the solution from a cold start is 2 mini seconds (ms). From a warm start, the execution time returned by *system_clock* is zero because the resolution of *system_clock* is one ms, and the time used by G-POLAR is less than 1 ms. In addition, the code is much simpler with the full numerical approach, and the convergence is more robust. Furthermore, since the algorithm is not limited to only the situations where the analytical solutions for the trajectory are available, greater applicability and flexibility are afforded to the algorithm. In fact, it is even not necessary to rely on the approximate switching function $\hat{K}$ in Eq. (61). See the next section on these points.

The propellant penalty due to the smoothing approximation in Eq. (61) to the optimal bang-bang control is negligible, on the order of 0.1% or less. An added practical advantage of the continuous thrust profile by Eq. (61) is that such a profile is more likely to be produced closely by the actual thrust of an engine than the bang-bang thrust. Figure 5 illustrates the engine throttle profiles in the same fuel-optimal powered descent problem, one solved with the smoothed thrust, and the other with the optimal bang-bang thrust. The parameter ε in Eq. (61) can be adjusted to affect how fast the thrust will change from one bound to the other; the closer the value of ε is to 1, the faster the response is, and vice versa.

The default solution setting in G-POLAR is the smoothed-thrust approach, while the user has the option to use the analytical equation based approach as outlined in Section IV.

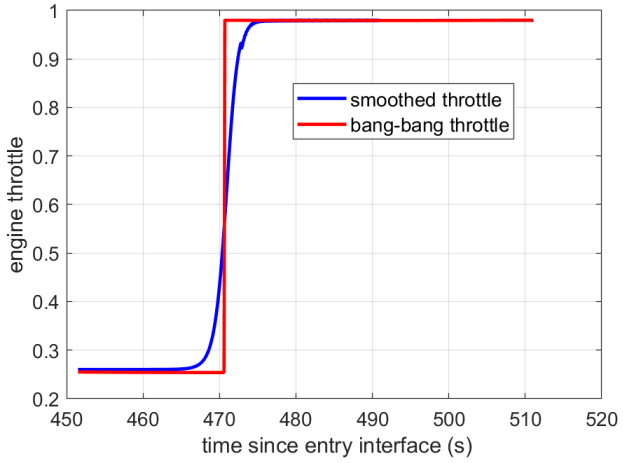


Fig. 5 Smoothed engine throttle and bang-bang throttle in the same case

VIII. Pointing Constraint

In the literature another inequality constraint that may be imposed in the fuel-optimal powered descent problem is the so-called “pointing constraint" [9] that limits the thrust vector direction deviation from the local vertical direction. Since the thrust direction defines (at least largely) the body attitude of the vehicle, this constraint effectively serves as a proxy for a body orientation constraint in a 6-degree-of-freedom setting. Such a constraint may arise because of a variety of operational reasons. In our notation this constraint may be expressed as

$$\cos\Theta - \mathbf{1}_T^T \mathbf{1}_r \leq 0 \tag{62}$$

where $\Theta > 0$ is the given half apex angle of a pointing cone within which the thrust vector must lie. See Fig. 6 for an illustration. The constant vector $\mathbf{1}_r$ is the unit vector in the vertical (up) direction at the landing site. The above constraint dictates that the angle between the thrust vector and vertical direction must be no greater than Θ. Since $\mathbf{1}_T$ defines the body attitude of the lander, this constraint effectively limits the vehicle attitude even though only point-mass dynamics are used. Note that when $\mathbf{1}_T$ is used as control, the pointing constraint (62) is always convex (linear) for any $0 \leq \Theta \leq \pi$, whereas it is a non-convex constraint when $\pi/2 < \Theta < \pi$ if the entire thrust vector $\boldsymbol{T} = T\mathbf{1}_T$ is used as the control [9], even though convexity is not required in our current approach.

A. Analysis of Optimal Thrust Direction

Constraint (62) limits the thrust vector inside a pointing cone with a half-apex-angle of $\Theta > 0$, as shown in Fig. 6. In the presence of the pointing constraint, the expression of the Hamiltonian H remains the same as in Eq. (15). The solutions for the costate vectors $\boldsymbol{p}_r$ and $\boldsymbol{p}_V$ are unchanged, still given by Eq. (16). Note that the only term dependent on $\mathbf{1}_T$ in the Hamiltonian is $(T/m)\boldsymbol{p}_V^T\mathbf{1}_T$. The Maximum Principle now calls for the optimal $\mathbf{1}_T^*$ to solve the following

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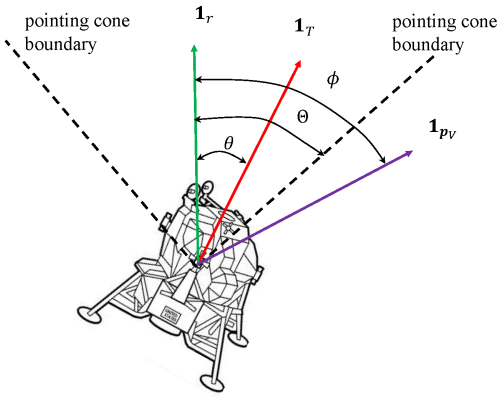


Fig. 6 Pointing constraint and relationship among the vertical direction $\mathbf{1}_r$, thrust direction $\mathbf{1}_T$, and primer vector direction $\mathbf{1}_{\boldsymbol{p}_V}$

pointwise finite-dimension optimization problem at each fixed $t \in [t_0, t_f]$ [12]

$$\max_{\mathbf{1}_T} \ \boldsymbol{p}_V^T \mathbf{1}_T \tag{63}$$

$$\mathbf{1}_T^T \mathbf{1}_T = 1 \tag{64}$$

$$\cos\Theta - \mathbf{1}_r^T \mathbf{1}_T \le 0 \tag{65}$$

Define

$$\phi = \cos^{-1}\left(\mathbf{1}_{\boldsymbol{p}_V}^T \mathbf{1}_r\right) \tag{66}$$

where $\mathbf{1}_{\boldsymbol{p}_V} = \boldsymbol{p}_V / \|\boldsymbol{p}_V\|$. Clearly ϕ is the angle between $\mathbf{1}_r$ and $\mathbf{1}_{\boldsymbol{p}_V}$. See Fig. 6 for a geometric illustration. Apply the Karush-Kuhn-Tucker (KKT) condition to the nonlinear programming problem in Eqs. (63)–(65), and consider the case when the constraint (65) is active, that is, $\phi > \Theta$ (the inactive case has been dealt with in Section III). In an active case it is necessary that $\phi \neq 0$. It can be shown that for $\phi \neq 0$, the above pointwise optimization problem has the solution $\mathbf{1}_T^*$ that lies in the plane formed by $\mathbf{1}_r$ and $\mathbf{1}_{\boldsymbol{p}_V}$, and is given by

$$\mathbf{1}_T^* = \frac{\sin(\phi - \Theta)}{\sin\phi}\mathbf{1}_r + \frac{\sin\Theta}{\sin\phi}\mathbf{1}_{\boldsymbol{p}_V} \tag{67}$$

Now $\mathbf{1}_T^*$ is no longer completely in the direction of $\mathbf{1}_{\boldsymbol{p}_V}$ as in Eq. (17), but has a component in the direction of $\mathbf{1}_r$ as well. The geometric interpretation is that in this case $\phi > \Theta$ as in Fig. 6, thus the control vector $\mathbf{1}_T^*$ cannot take the direction of just $\mathbf{1}_{\boldsymbol{p}_V}$, because it will violate the pointing limit (65). The optimal solution now is one where $\mathbf{1}_T^*$ is exactly at the intersection between the surface of the pointing cone and the plane of $\mathbf{1}_r$ and $\mathbf{1}_{\boldsymbol{p}_V}$.

Consider a thrust direction $\mathbf{1}_T$ that is in the plane formed by $\mathbf{1}_r$ and $\mathbf{1}_{p_V}$, but otherwise in an arbitrary direction. Let θ be the angle between this $\mathbf{1}_T$ and $\mathbf{1}_r$ (cf. Fig. 6). Then it is easy to show, based on the geometry in Fig. 6, that

$$\mathbf{1}_T = \frac{\sin(\phi - \theta)}{\sin\phi}\mathbf{1}_r + \frac{\sin\theta}{\sin\phi}\mathbf{1}_{\boldsymbol{p}_V} \tag{68}$$

The optimal thrust vector $\mathbf{1}_T^*$, in both cases when the constraint (65) is inactive and active, can be defined by a single equation (68), if θ is evaluated in the following way

$$\theta = \begin{cases} \phi, & \text{if } \phi \le \Theta \\ \Theta, & \text{if } \phi > \Theta \end{cases} \tag{69}$$

The first case in Eq. (69) corresponds to the case where constraint (65) is inactive, and Eq. (68) with $\theta = \phi$ gives the solution in Eq. (17); The 2nd case in Eq. (69) is the case where constraint (65) is active, and Eq. (68) with $\theta = \Theta$ gives the solution in Eq. (67).

This development leads to the following general result:

Conclusion:

The optimal thrust direction $\mathbf{1}_T^*$ in the fuel-optimal powered descent problem with the pointing constraint (65) is given by Eq. (68) with θ defined in Eq. (69); this solution is valid regardless of whether the pointing constraint is inactive or active.

The advantage of defining the optimal thrust direction in the unifying way in Eqs. (65) and (69) is that along the trajectory as ϕ varies continuously, the optimal thrust direction is found accordingly. There is no need to explicitly determine when and whether the constraint (62) is active or not, and for how many times. This benefit significantly reduces the complexity of the algorithm when the indirect method is used.

B. Unifying Method of Solution

Another major technical complication in the presence of an active pointing constraint (62) is that the optimal thrust direction in Eq. (68) will also change the optimal thrust magnitude profile. The switching function in this case is now

$$S = \frac{\boldsymbol{p}_V^T \mathbf{1}_T^*}{m} - \frac{p_m}{V_e} = \frac{1}{m}\left[\frac{\sin(\phi-\theta)}{\tan\phi} + \frac{\sin\theta}{\sin\phi}\right] p_V - \frac{p_m}{V_e} \tag{70}$$

where $p_V = \sqrt{p(t-t_0)^2 + q(t-t_0) + r}$ still, but the governing equation of p_m is now changed to

$$\dot{p}_m = \frac{T^* \boldsymbol{p}_V^T \mathbf{1}_T^*}{m^2} = \frac{T^*}{m^2}\left[\frac{\sin(\phi-\theta)}{\tan\phi} + \frac{\sin\theta}{\sin\phi}\right] p_V \tag{71}$$

Note that Eqs. (70) and (71) are valid regardless of whether or not the pointing constraint is active, as long as θ is defined by Eq. (69). It remains an open question whether an approximation to the switching function can be found when the pointing constraint is active somewhere in $[0, t_f]$, similar to that in Ref. [6] for the case without the pointing constraint. Such an approximation may help make the problem simpler, but it is likely difficult to find, if possible at all.

Yet with the smoothed-thrust setting presented in Section VII, and the unifying expressions in Eqs. (68), (69), (70) and (71) derived here, there is neither conceptual nor numerical difficulty to solve this problem. We only need to replace K_c in the unknown vector $\boldsymbol{z}$ of the root-finding problem (44) by the initial costate $p_m(0)$, and replace the constraint in Eqs. (37) and (43) with the transversality condition on $p_m(t_f)$ from Eq. (27):

$$p_m(t_f) - 1 = 0 \tag{72}$$

For such a given $\boldsymbol{z} \in R^8$, the state equations in (6) and the mass costate equation Eq. (71) are numerically integrated forward from 0 to t_f with $\mathbf{1}_T^*$ given by Eqs. (62) and (74). Along the way the switching function $S(t)$ is calculated from Eq. (70) at each $t \in [0, t_f]$, and then used in Eq. (61) to replace $\hat{K}(t)$, which in turn defines $T^*(t)$ at each $t \in [0, t_f]$

$$T^*(t) = \frac{1}{2}(T_{max} + T_{min}) + \frac{1}{2}(T_{max} - T_{min}) \tanh\left[\frac{S(t)}{1-\varepsilon}\right], \quad 0 < \varepsilon < 1 \tag{73}$$

A gradient-based method can be used to update $\boldsymbol{z}$ to solve the problem the same way as outlined in Section VII. Note that the piecewise near constancy of T^* from Eq. (73) will not cause any vanishing sensitivity issues for a gradient-based method in solving the problem. The technical reasoning would take more space and is omitted here.

The need for numerically integrating $p_m(t)$ and satisfying the constraint in Eq. (72) poses no convergence difficulty because $p_m(t)$ is monotonically increasing by Eq. (71) (recall that $\phi - \theta \geq 0$ by Eq. (69)), and $\dot{p}_m$ is bounded for $\phi \neq 0$ (and $m > 0$). That is, $p_m(t)$ is well behaved and an arbitrary initial guess of $p_m(0) \in (0, 1)$ is sufficiently good. Indeed, the discussion also reveals that even if no pointing constraint is imposed, the solution approach based on the smoothed-thrust setting in G-POLAR need not rely on the approximate switching function $\hat{K}$ but directly computing p_m and using the true switching function S, just like what is described above when a pointing constraint is imposed.

As a final algorithmic refinement, recall that Θ is specified in the problem statement and ϕ is part of the solution that varies along the trajectory as $\boldsymbol{p}_V$ varies (thus ϕ is dependent on the unknown $\boldsymbol{z}$). Thus Eq. (69) may be regarded as that θ is the output of a saturation function whose input is ϕ with the limit of the linear region at $\phi = \Theta$. This is a continuous function, but non-differentiable at $\phi = \Theta$. To eliminate the possibility that this non-smoothness may again cause potential convergence difficulty when the problem is solved by a gradient-based method in a shooting setting like

in this paper, the following L^p-norm-based differentiable sigmoid function may be used to closely approximate the saturation function in (69):

$$\theta = \Theta \left[\frac{(\phi/\Theta)}{\sqrt[p]{1 + (\phi/\Theta)^p}} \right] \tag{74}$$

where $p \geq 1$ is a constant. When p is chosen to be sufficiently large (for instance, $p = 100$), Eq. (74) is a very good approximation to the saturation function in Eq. (69), but it is continuously differentiable everywhere. Any small inaccuracy in approximating Eq. (69) by Eq. (74), mostly around $\phi/\Theta = 1$, is not critical at all (hence no need for a more elaborate approximation, such as something similar to Eq. (56)).

One should realize that a pointing constraint with a restrictive Θ will severely limit what trajectory the lander can fly as well as what PDI condition is feasible for landing. If the objective is to ensure that the pitch angle at touchdown is such that the vehicle has the correct attitude (e. g., vertical for a vertical landing vehicle), adding a terminal descent phase guided by a different guidance scheme (fractional-polynomial powered descent guidance in Ref. [14] for instance) after fuel-optimal powered descent is a much more preferable alternative. See Ref. [11] for such an approach.

Figure 7 shows the comparison of pitch angle profiles in the same powered descent mission at Mars with and without a pointing constraint. The pitch angle in this case is defined to be the angle between the thrust vector and the local horizontal plane at the landing site. The vehicle data and mission scenario will be described in Section X. When a pointing constraint with $\Theta = 75$ deg is imposed, the pitch angle is limited to a value of $90 - \Theta = 15$ deg for an initial period of time, which is automatically determined by G-POLAR. Note that the final pitch angle is not 90 degrees. This is because in the point-mass formulation the thrust direction (which represents the body attitude) is the control which is to be determined by the optimal solution. The final thrust direction cannot be readily constrained in a 3DOF optimal control formulation. However, in most landing missions there will be a terminal vertical descent phase before touchdown guided by more conventional method (see Ref. [4]). The fuel-optimal guidance can deliver the vehicle to the start of the terminal descent phase. The terminal phase guidance will ensure that final pitch attitude is 90 degrees, even if the pitch attitude at the start of the terminal phase is not exactly vertical [4].

As expected, the pointing constraint has a visible effect on the trajectory as seen in Fig. 8, particularly in the altitude profiles. Both cases meet the specified landing condition accurately. What is less intuitive but not difficult to comprehend is the influence of the pointing constraint on the optimal thrust magnitude profile, plotted in Fig. 9. The presence of the active pointing constraint alters noticeably when the optimal thrust switches in the bang-bang structure. Despite the visible differences in the trajectories, the propellant usage increases by just 1.09% when the pointing constraint is imposed, a testimony to the fact that the (constrained) optimality is still retained when the pointing constraint is imposed and active.

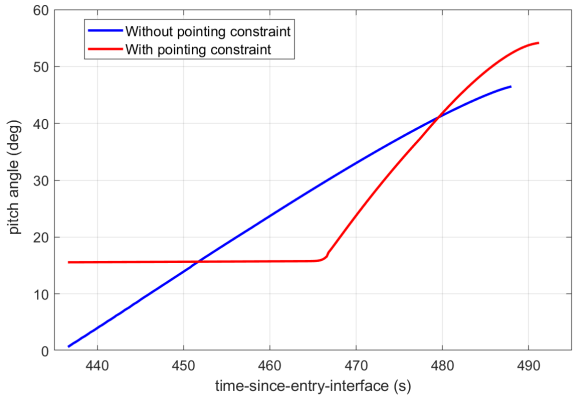


Fig. 7 Pitch angle profiles with and without pointing constraint

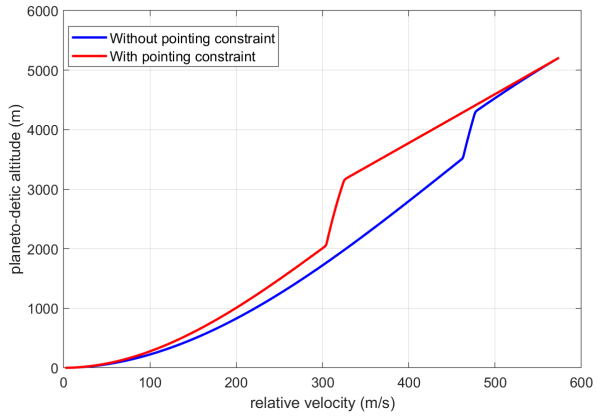


Fig. 8 Powered descent trajectories with and without pointing constraint

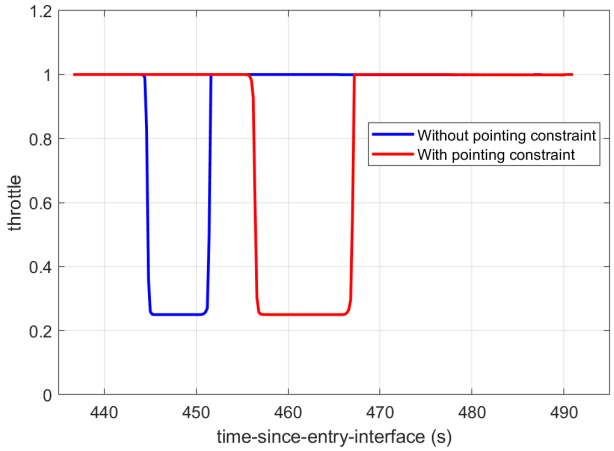


Fig. 9 Optimal throttle profiles with and without pointing constraint in the same mission

IX. Tracking Guidance

A. What Could Go Wrong

When an algorithm for the fuel-optimal powered descent problem is deployed onboard a vehicle, in each guidance cycle the current state is used as the initial condition. The commanded thrust magnitude and the thrust unit vector at the current time from the obtained optimal solution are the current guidance commands. The dependence of the guidance commands on the current actual state renders the guidance solution effectively closed loop if the guidance update frequency is sufficiently high.

The conventional wisdom is to continue this process in every guidance cycle until the touchdown (or arrival at an otherwise specified final condition). In the soft-landing powered descent problem and many other different types of guidance problems this strategy works just fine. But something peculiar in the fuel-optimal pinpoint-landing problem requires special attention. Due to the fully constrained final condition in the pinpoint-landing solution, in some cases trajectory dispersion could cause a phenomenon we call “solution degradation" toward the end of the descent. The normal fuel-optimal solution is expected to end at the maximum thrust [4], and the thrust direction will change only relatively gently. Yet as the vehicle gets closer to the landing site along the fuel-optimal descent trajectory, unfavorable trajectory dispersion, even if small, could at some point put the condition in such a state where very rapid changes of the thrust direction, and sometimes an unanticipated extra thrust magnitude switch to the minimum throttle, are needed in

order to satisfy the 6 position and velocity constraints.

Figure 10 shows a representative case. This is a solution found by the algorithm in closed-loop simulation when time-since-PDI reaches 27 seconds. In this solution at about 3 seconds before touchdown, the 3 thrust acceleration components in the converged pinpoint-landing solution start having dramatic and fast changes. Indeed. the vertical component of the thrust vector (a_{T_y} in the figure) not only doubles its magnitude in a matter of 3 seconds, but ends at the opposite sign! All the while, the total thrust magnitude remains at the constant, maximum level as required by the fuel optimality, despite the large and rapid changes in the components of the thrust vector near the end. Assuming that the thrust vector is aligned with the longitudinal body axis, these large changes in the thrust components within the last 3 seconds mean that the attitude of the vehicle will go through rapidly and large change. Indeed, Fig. 11 confirms that the pitch angle rapidly decreases in the last 3 seconds to near -80 deg. The vehicle effectively will land upside down by flying this trajectory. Even though the trajectory with the thrust acceleration in Fig. 10 satisfies all the imposed landing constraints on the problem, and all the necessary conditions for fuel optimality, it is simply not acceptable to any lander. Attempt to prevent the pitch from becoming negative by tightening the pointing constraint (62) would result in the problem having no solution in this case.

This phenomenon is inherent to the fuel-optimal pinpoint-landing problem when the remaining flight time is relatively short, because of its exceedingly constrained nature – fully constrained final state with zero variation in the thrust magnitude. In the late part of the powered descent when T^* is fixed at T_{max}, the only way to control the full trajectory state, both position and velocity, is through the direction of the thrust vector. When the time-to-go becomes shorter, trajectory dispersion can and will in some cases render the type of unrealistic maneuvers in Fig. 10 in the solution to the fuel-optimal pinpoint-landing problem. In general, such a solution only happens after t_2, the instant of the second switch of the bang-bang thrust (see Fig. 2) to T_{max}. The reason is that before t_2, the problem still has the degree-of-freedom to adjust t_2, which effectively provides a way to modulate the average thrust pulse as an additional means to control the velocity. After t_2, the trajectory loses this leverage.

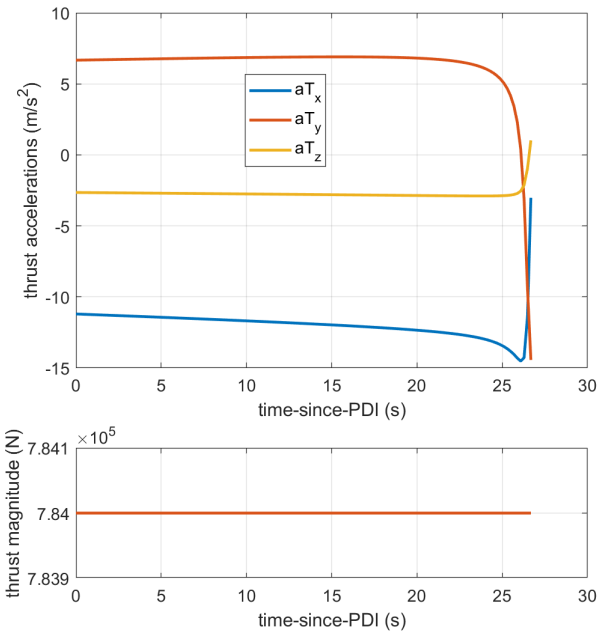


Fig. 10 Rapid changes in thrust acceleration components in a North-Up-East topocentric frame near the end of fuel-optimal pinpoint-landing trajectory

On the other hand, the fuel-optimal soft-landing problem and close-approach problem with a small weighting coefficient are free from this phenomenon. This is because the landing location of the trajectory in these problems is not tightened down to a single point. The trajectory always has the freedom to move the landing location around to accommodate dispersions, without the need to make drastic maneuvers.

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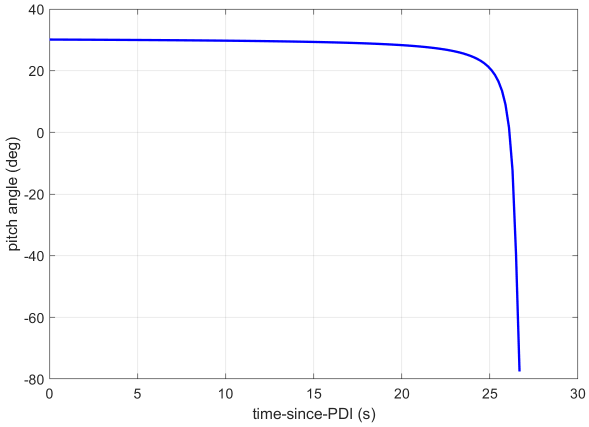


Fig. 11 Pitch angle corresponding to the thrust direction in Fig. 10

B. Tracking Guidance Law

To mitigate the types of serious potential issues seen in Figs. 10 and 11 in the fuel-optimal pinpoint-landing problem in the later part of the powered descent, and still take advantage of the capability of fuel-optimal guidance, an approach is to use the fuel-optimal solution for guidance for the first part of the powered descent of a pinpoint-landing mission. Because the solution provides the full information of the complete optimal trajectory, the trajectory is stored as the reference in preparation for tracking. At an appropriate time in the later part of the powered descent (more on this later), the guidance is switched to the tracking mode where a tracking law is deployed to follow the last converged optimal solution. The tracking law will always be able to generate guidance commands reliably, and if it is effective, the actual trajectory will be practically the same as the converged optimal solution, therefore ensuring the predictability of the trajectory and accurate satisfaction of the touchdown condition.

To develop the tracking law, we will begin by representing the 3-DOF dynamics of our problem in the following fashion

$$\dot{\boldsymbol{r}} = \boldsymbol{V} \tag{75}$$

$$\dot{\boldsymbol{V}} = \boldsymbol{g} + \boldsymbol{u} \tag{76}$$

where $\boldsymbol{r} = (x\ y\ z)^T$ is the position vector, $\boldsymbol{V} = (u\ v\ w)^T$ the velocity vector, and $\boldsymbol{u} = \boldsymbol{a}_T = \boldsymbol{T}/m$ the thrust acceleration vector, treated as the trajectory control. Let $\boldsymbol{r}^*(t)$, $\boldsymbol{V}^*(t)$, and $\boldsymbol{u}^*(t)$, $0 \le t \le t_f^*$ (where $t = 0$ means the current instant), be the reference state and control profiles, obtained from a converged solution of the pinpoint-landing problem. Because the optimal solution is dynamically feasible, the reference profiles satisfy the same dynamics

$$\dot{\boldsymbol{r}}^* = \boldsymbol{V}^* \tag{77}$$

$$\dot{\boldsymbol{V}}^* = \boldsymbol{g} + \boldsymbol{u}^* \tag{78}$$

We want to find a feedback tracking law for $\boldsymbol{u}$ so that $\boldsymbol{r} \to \boldsymbol{r}^*$, and $\boldsymbol{V} \to \boldsymbol{V}^*$ asymptotically for all possible (and reasonable) dispersion in the initial condition of the actual trajectory. Unlike a stabilization problem or an output-tracking problem, the challenges in this tracking problem rest with the fact that we wish to achieve full-state tracking (position and velocity) of a time-varying reference profile (that is in general not a polynomial of time). In the following a simple and effective tracking guidance law is developed based on the predictive control method in Ref. [10] for our purpose.

Define the tracking errors

$$\boldsymbol{e}_1(t) = \boldsymbol{r}(t) - \boldsymbol{r}^*(t) \tag{79}$$

$$\boldsymbol{e}_2(t) = \boldsymbol{V}(t) - \boldsymbol{V}^*(t) \tag{80}$$

Following the one-step-ahead predictive control method in Ref. [10], we seek to minimize the following tracking error function at any $t \in [0, t_f)$ and for a horizon $h > 0$

$$\min_{\boldsymbol{u}(t)} J = \frac{1}{2}\boldsymbol{e}_1^T(t+h)^T Q_1 \boldsymbol{e}_1(t+h) + \frac{1}{2}\boldsymbol{e}_2(t+h)^T Q_2 \boldsymbol{e}_2(t+h) \tag{81}$$

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where Q_1 and Q_2 are 3×3 positive definite weighting matrices. We will proceed to predict $\boldsymbol{e}_1(t+h)$ as a function of $\boldsymbol{u}(t)$ by a 2nd-order Taylor series at t, and $\boldsymbol{e}_2(t+h)$ a 1st-order Taylor series at t:

$$\boldsymbol{e}_1(t+h) \approx \boldsymbol{e}_1(t) + h\dot{\boldsymbol{e}}_1(t) + \frac{h^2}{2}\ddot{\boldsymbol{e}}_1(t) = \boldsymbol{e}_1(t) + h\dot{\boldsymbol{e}}_1(t) + \frac{h^2}{2}\left[\boldsymbol{g} + \boldsymbol{u}(t) - \dot{\boldsymbol{V}}^*(t)\right] \tag{82}$$

$$\boldsymbol{e}_2(t+h) \approx \boldsymbol{e}_2(t) + h\dot{\boldsymbol{e}}_2(t) = \boldsymbol{e}_2(t) + h\left[\boldsymbol{g} + \boldsymbol{u}(t) - \dot{\boldsymbol{V}}^*(t)\right] \tag{83}$$

Using the approximations in Eqs. (82) and (83) in Eq. (81) leads to a convex quadratic function of $\boldsymbol{u}(t)$. The unique minimizing $\boldsymbol{u}(t)$ is found from $\partial J/\partial \boldsymbol{u}(t) = 0$

$$\boldsymbol{u}(t) = -P\left\{\frac{1}{2h^2}Q_1\left[\boldsymbol{e}_1(t) + h\dot{\boldsymbol{e}}_1(t) + \frac{h^2}{2}(\boldsymbol{g} - \dot{\boldsymbol{V}}^*(t))\right] + \frac{1}{h}Q_2\left[\boldsymbol{e}_2(t) + h(\boldsymbol{g} - \dot{\boldsymbol{V}}^*(t))\right]\right\} \tag{84}$$

where P is the positive definite matrix

$$P = \left(\frac{1}{4}Q_1 + Q_2\right)^{-1} \tag{85}$$

Since $\dot{\boldsymbol{V}}^* = \boldsymbol{g} + \boldsymbol{u}^*$, replacing $(\boldsymbol{g} - \dot{\boldsymbol{V}}^*)$ by $-\boldsymbol{u}^*$ in the expression of $\boldsymbol{u}(t)$ and simplifying yield

$$\boldsymbol{u}(t) = \boldsymbol{u}^*(t) - P\left[\frac{1}{2h^2}Q_1\boldsymbol{e}_1 + \frac{1}{h}\left(\frac{1}{2}Q_1 + Q_2\right)\boldsymbol{e}_2\right] \tag{86}$$

Let us choose for simplicity

$$Q_1 = k_1 I_{3\times 3}, \quad Q_2 = k_2 I_{3\times 3}, \quad k_1 > 0,\ k_2 > 0 \tag{87}$$

Then

$$\boldsymbol{u}(t) = \boldsymbol{u}^*(t) - \frac{1}{2h^2}\left(\frac{k_1}{k_1/4 + k_2}\right)\boldsymbol{e}_1 - \frac{1}{h}\left(\frac{k_1/2 + k_2}{k_1/4 + k_2}\right)\boldsymbol{e}_2 \tag{88}$$

Define

$$k = \frac{2}{1 + 4k_2/k_1} \tag{89}$$

Clearly for $k_1 > 0$ and $k_2 > 0$

$$0 < k < 2 \tag{90}$$

The control law for $\boldsymbol{u}(t)$ can be further simplified to be

$$\boldsymbol{u}(t) = \boldsymbol{u}^*(t) - \frac{k}{h^2}\boldsymbol{e}_1 - \frac{1}{h}\left(\frac{1}{2}k + 1\right)\boldsymbol{e}_2 \tag{91}$$

Denote

$$k_r = k, \quad k_v = \frac{k}{2} + 1 \tag{92}$$

Replacing $\boldsymbol{u}$, $\boldsymbol{u}^*$, $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ with their definitions, the tracking control law takes the final form

$$\boldsymbol{a}_T(t) = \boldsymbol{a}_T^*(t) - \frac{k_r}{h^2}\left[\boldsymbol{r}(t) - \boldsymbol{r}^*(t)\right] - \frac{k_v}{h}\left[\boldsymbol{V}(t) - \boldsymbol{V}^*(t)\right], \quad k_v = k_r/2 + 1 \tag{93}$$

There are two controller parameters that define the controller gains

$$0 < k_r < 2, \quad h > 0 \tag{94}$$

It is rather interesting to note that a tracking control law for the powered descent problem may be constructed based on a very different approach that is closely related to the Apollo lunar powered descent guidance law (see Eqs. (12) and (41) in Ref. [14])

$$\boldsymbol{a}_T(t) = \boldsymbol{a}_T^*(t) - \frac{k_r}{t_{go}^2}\left[\boldsymbol{r}(t) - \boldsymbol{r}^*(t)\right] - \frac{k_v}{t_{go}}\left[\boldsymbol{V}(t) - \boldsymbol{V}^*(t)\right], \tag{95}$$

where $t_{go} = t_f - t$ and $k_r > 0$, and k_v may be selected from (Eq. (42) in Ref. [14])

$$k_v = \frac{k_r}{\gamma + 2} + \gamma + 1 \tag{96}$$

where $\gamma \geq 0$ is a constant. When $\gamma = 1$ and $k_r = 12$, and $\boldsymbol{a}_T^*(t)$ is a quadratic function of time, the guidance law in Eq. (95) is the Apollo lunar descent guidance law [15]; When $\gamma = 1$ and $k_r = 6$, and $\boldsymbol{a}_T^*(t)$ is a linear function of time, the guidance law in Eq. (95) becomes the E-guidance law [16], another Apollo era powered descent guidance law. Remarkably, the E-guidance law later is found to be the solution to an optimal powered descent problem with a quadratic performance index [17]. Clearly when $\gamma = 0$, the pair of k_r and k_v in Eq. (96) satisfy the same relationship as in Eq. (93). The difference between tracking laws in Eqs. (93) and (95) is that the time-to-go t_{go} in Eq. (95) is replaced by a user-selected constant h in Eq. (93). However, guidance law (93) will be shown to have guaranteed asymptotic tracking of any feasible reference trajectory defined by an arbitrary feasible $\boldsymbol{a}_T^*(t)$, not just linear or quadratic functions of time.

C. Asymptotic Tracking Guarantee

We want to make sure that, under the derived predictive control law in the general form in Eq. (86) for any positive definite Q_1 and Q_2 (thus including the special case in Eq. (87) that is re-arranged in the form of Eq. (93)), we have a theoretic guarantee that $\boldsymbol{e}_1 \to 0$ and $\boldsymbol{e}_2 \to 0$ globally. It can easily be shown that the closed-loop tracking error dynamics with the control law (86) are

$$\dot{\boldsymbol{e}}_1 = \boldsymbol{e}_2 \tag{97}$$

$$\dot{\boldsymbol{e}}_2 = -\frac{1}{2h^2} P Q_1 \boldsymbol{e}_1 - \frac{1}{h} P \left(\frac{1}{2} Q_1 + Q_2 \right) \boldsymbol{e}_2 \tag{98}$$

Consider the Lyapunov function candidate [10]:

$$V = \frac{1}{4h} \boldsymbol{e}_1^T Q_1 \boldsymbol{e}_1 + \frac{1}{2h} \boldsymbol{e}_2^T P^{-1} \boldsymbol{e}_2 \tag{99}$$

This $V > 0$ for any $\boldsymbol{e} = (\boldsymbol{e}_1^T \ \boldsymbol{e}_2^T)^T \neq 0$, and $V = 0$ when $\boldsymbol{e} = 0$. Furthermore,

$$\dot{V} = -\boldsymbol{e}_2^T \left(\frac{1}{2} Q_1 + Q_2 \right) \boldsymbol{e}_2 \leq 0 \tag{100}$$

It is evident that $\dot{V} < 0$ whenever $\boldsymbol{e}_2 \neq 0$. Therefore V will continue to decrease until $\boldsymbol{e}_2 = 0$. From Eq. (98), $\boldsymbol{e}_2 \equiv 0$ necessitates that

$$\frac{1}{2h^2} P Q_1 \boldsymbol{e}_1 = 0 \tag{101}$$

Since $0 < h < \infty$, the above condition means $P Q_1 \boldsymbol{e}_1 = 0$. By LaSalle's Invariance Principle [18], the total tracking error $\boldsymbol{e}$ will asymptotically approach the following invariant set $\mathcal{S}_V$ where $\dot{V} = 0$

$$\mathcal{S}_V = \left\{ \boldsymbol{e} = (\boldsymbol{e}_1^T \ \boldsymbol{e}_2^T)^T \,\middle|\, \boldsymbol{e}_2 = 0, \ P Q_1 \boldsymbol{e}_1 = 0 \right\} \tag{102}$$

Since P and Q_1 both are positive definite, PQ_1 is full rank. The condition $PQ_1\boldsymbol{e}_1 = 0$ has a unique solution of $\boldsymbol{e}_1 = 0$. Hence, the invariant set $\mathcal{S}_V$ has only one point of $\boldsymbol{e} = 0$. Thus, under the control law in Eq. (86), or its special case in Eq. (93), the tracking error $\boldsymbol{e} \to 0$ globally.

It is noteworthy that even though truncated Taylor series expansions are used to approximate the predicted tracking errors in Eqs. (82) and (83), the above discussion shows that global tracking stability requires no assumption on the smallness of the prediction horizon h, a somewhat counter-intuitive pleasant surprise. This finding leaves the selection of h unencumbered by stability concerns, free to be adjusted for tracking performance considerations as seen in the following.

Now with asymptotic tracking ensured, we will briefly examine the effects of the parameters k (which is the same as k_r) and h in (91) on tracking performance. It is easy to show that the closed-loop tracking error dynamics under control law (91) are equivalent to

$$\ddot{\boldsymbol{e}}_1 + \frac{(k/2+1)}{h} \dot{\boldsymbol{e}}_1 + \frac{k}{h^2} \boldsymbol{e}_1 = 0 \tag{103}$$

The damping ratio and natural frequency of this second-order system are

$$\zeta = \frac{(k/2+1)}{2\sqrt{k}} \tag{104}$$

$$\omega_n = \frac{\sqrt{k}}{h} \tag{105}$$

Interestingly, the damping ratio is just a function of k, independent of h. Since $0 < k < 2$, it can be further ascertained that the closed-loop system is overdamped when $0 < k < 6 - 4\sqrt{2} = 0.3431$, and underdamped when $0.3431 < k < 2$. For instance, $k = 2 - \sqrt{2} = 0.5858$ will result in $\zeta = \sqrt{2}/2 = 0.707$. On the other hand, the natural frequency, or the bandwidth, of the closed-loop system is inversely proportional to $h > 0$. A smaller h produces larger tracking gains and tighter tracking, and a larger h reduces the tracking gains and lessens the possible control saturation. These conclusions can help with the selection of the values for k and h. Our experience shows that good tracking performance is not at all sensitive to the choices of k and h. For example, for a high-mass (about 60 metric tons) Mars entry, descent and landing (EDL) mission with a supersonic retropropulsion system, the powered descent phase lasts 50-70 seconds, and the tracking guidance may be deployed in the last 30 seconds. The following (more or less arbitrary) choices of tracking gains work very well as many other choices

$$k = 2 - \sqrt{2} = 0.5858 \tag{106}$$

$$h = 1.0 \quad \text{(s)} \tag{107}$$

These values give a damping ratio of $\zeta = 0.707$ and natural frequency of $\omega_n = 0.765$, resulting in an estimated settling time of the closed-loop tracking of $4.6/\zeta\omega_n = 8.5$ seconds.

It is recommended that the tracking law be deployed before the solution degradation can potentially occur to ensure the quality of the reference trajectory. Therefore the latest timing should be shortly after t_2, the second switch of the bang-bang thrust magnitude profile. G-POLAR allows the user to choose the tracking option and it is recommended for the pinpoint-landing problem. Once the tracking option is selected, G-POLAR will automatically determine the time-to-go to deploy the tracking guidance.

To leave a margin in the thrust magnitude for the tracking law to modulate the throttle for tracking the fuel-optimal solution, an upper thrust bound T_{max} slightly smaller than the full thrust of the engine, and a lower bound T_{min} slightly larger than the actual lower bound should be used in the generation of the fuel-optimal solution. This margin will come at the cost of some penalty of propellant consumption as the fuel-optimal solution calls for the maximum and minimum thrust available. How much this margin should be is a classical trade off between performance and robustness with respect to trajectory dispersion expected in the tracking phase. In our Monte Carlo testing in the next section, a 2% margin appears sufficient and the propellant penalty is about 1.5 – 2%.

No tracking guidance would be necessary for soft-landing problem or close-approach problem with a small to medium weighting coefficient $\kappa > 0$. The fuel-optimal guidance solution will be well behaved to the end.

X. Evaluation and Testing

A. Independent Verification

To verify that the solution found by G-POLAR is indeed fuel optimal, a state-of-the-art pseudospectral optimal control software [19] is used to solve the identical problems, and the solutions are compared against each other. The pseudospectral optimal control method uses orthogonal polynomial based interpolations to discretize the state and costate equations, and the resulting nonlinear programming problem is solved on the basis of the KKT conditions. Therefore the pseudospectral optimal control software in Ref. [19] is independent of G-POLAR. To produce a true bang-bang solution, the knowledge that the optimal trajectory has at most 3 arcs of piecewise constant thrust is utilized, and the “knotting" technique in Ref. [20] is used. Comparison is performed over a range of different problems where the optimal thrust structures include all 3 possibilities of max-min-max, min-max, and max. The agreement of the solutions is excellent. Figure 12 shows the comparison of the throttle profiles for a problem with 3 burn arcs. The two methods generate essentially the same solutions. The comparison plots of other trajectory parameters are seen in Fig. 13.

The solutions with an active pointing constraint by G-POLAR are also verified by the same pseudospectral optimal control software. Figure 14 compares the pitch angle profiles in the presence of an active pointing constraint by G-POLAR and pseudospectral optimal solution. The case is the same as the one the constrained profile in Fig. 7 represents. Again the comparison is quite close. The propellant usages are practically the same in the two solutions (the difference is 0.01% or less). This independent verification establishes confidence that G-POLAR is indeed fuel-optimal, with or without the pointing constraint.

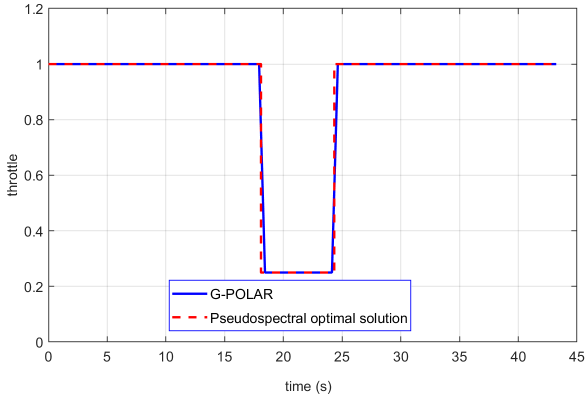


Fig. 12 Comparison of throttle profiles obtained by G-POLAR and pseudospectral optimal control software [19]

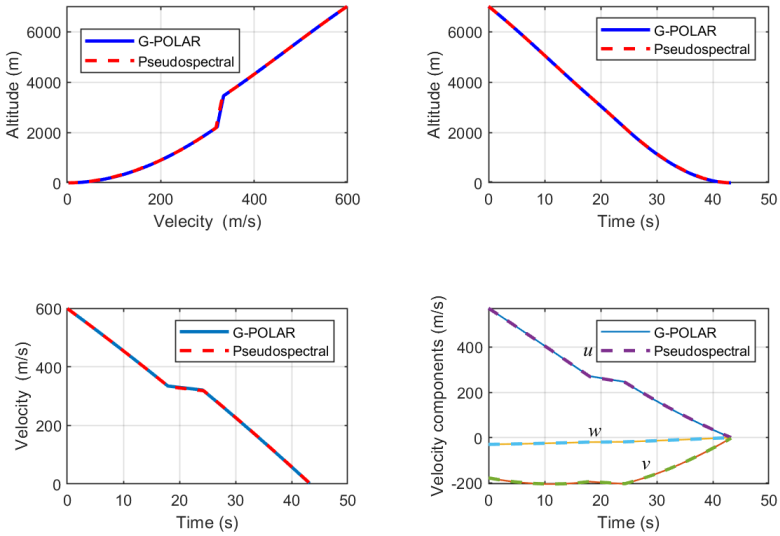


Fig. 13 Comparison of trajectory profiles obtained by G-POLAR and pseudospectral optimal control software [19]

B. Monte Carlo Closed-Loop Simulations

Stress testing of G-POLAR as a potential guidance algorithm is performed in closed-loop, 3DOF Monte Carlo simulations. The vehicle model used is that of a rigid mid lift-to-drag ratio entry vehicle concept for high-mass Mars EDL mission [21]. After the entry phase, the vehicle uses a supersonic retropropulsion system (SRP) to land horizontally. The closed-loop end-to-end simulation is performed from the entry interface (EI) to touchdown. The entry phase is guided by the Fully Numerical Predictor-corrector Entry Guidance (FNPEG) algorithm [22], and powered descent phase by G-POLAR in pinpoint-landing mode. The PDI point where the SRP is turned on and the powered descent phase starts is determined by the adaptive PDI logic presented in Ref. [4] that uses the fuel-optimal soft-landing solution to predict the PDI point. In the simulations here, G-POLAR is used in the soft-landing mode in this logic. This adaptive PDI logic determines a suitable PDI point based on the actual full state condition in each dispersed trajectory, there the PDI point can quite different from one dispersed trajectory to another.

The trajectory dispersion and uncertainty used in the Monte Carlo simulations include dispersion in EI state and

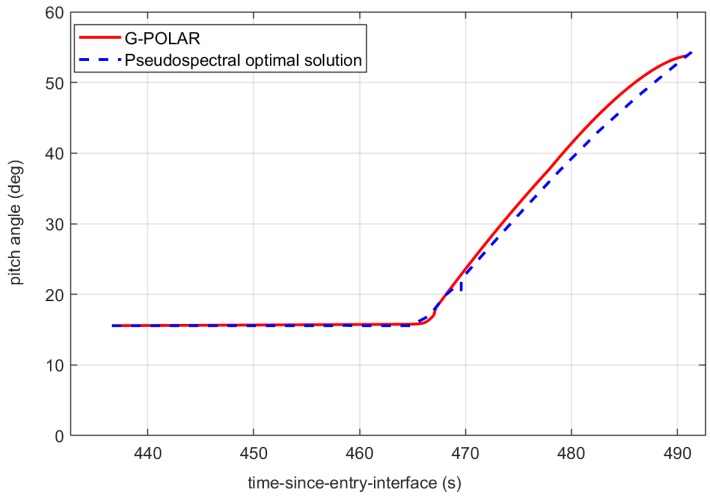


Fig. 14 Comparison of pitch angle profiles obtained by G-POLAR and pseudospectral optimal control software [19] with an active pointing constraint

initial mass, uncertainty in the aerodynamic lift and drag coefficients, Martian atmospheric density, and maximum thrust magnitude. The atmospheric density and its uncertainty are generated by the Mars Global Reference Atmospheric Model [23]. The dispersion and uncertainty in the entry phase directly impact the powered descent phase with different PDI condition. Table 1 summarizes the dispersion and uncertainty, and is similar to Table 1 in Ref. [24]:

Table 1 Dispersions for Monte Carlo Simulations

Dispersion	distribution	3σ/maximum	-3σ/minimum
EI planetodetic altitude (m)	Gaussian	300	-300
EI longitude (deg)	Gaussian	0.75	-0.75
EI planetodetic latitude (deg)	Gaussian	0.75	-0.75
EI inertial velocity (m/s)	Gaussian	10.0	-10.0
EI inertial flight-path angle (deg)	Gaussian	0.3	-0.3
EI inertial azimuth (deg)	Gaussian	0.5	-0.5
EI vehicle mass (percentage)	Gaussian	1.0%	-1.0%
C_L (percentage)	Uniform	10.0%	-10.0%
C_D (percentage)	Uniform	10.0%	-10.0%
Atmospheric density	Mars GRAM 2010	Mars GRAM 2010	Mars GRAM 2010
Mars GRAM 2010 *dusttau*	Uniform	0.9	0.1
Total SRP max thrust (percentage)	Gaussian	1.0%	-1.0%

The guidance frequency for FNPEG in the entry phase is 1 Hz, and 10 Hz for G-POLAR in the powered descent phase. G-POLAR guides the vehicle from PDI to touchdown – no terminal phase is used in the simulations. No navigation uncertainty is considered. In this sense the Monte Carlo simulations help establish the benchmark on the capability and precision of the guidance algorithm, given perfect navigation information. Each batch of Monte Carlo testing consists of a total of 3000 dispersed runs.

The required touchdown conditions are to land at the designated landing site, with a final velocity vertically downward at 2.5 m/s. No pointing constraint is imposed in the following results.

The dispersion and uncertainty in Table 1 generate widely different entry trajectories, consequently the PDI condition determined by the adaptive logic in Ref. [4] in each case can be significantly different. Figure 15 depicts the spread of PDI velocity and altitude. The PDI velocity ranges from 450 m/s to 570 m/s (Mach 1.8 to 2.4), and the PDI altitude can

be anywhere between 1.2 km and 3.4 km. The PDI ground range to landing site (not shown) is from 9.3 km to 13.8 km. Still, the adaptive PDI logic in Ref. [4] ensures that the PDI condition is feasible, despite the large variations. G-POLAR then takes the vehicle from these very different PDI conditions to touchdown at the landing site. The touchdown locations of the 3000 trajectories are shown in Fig. 16. This EDL mission is almost due north. The miss distances to the landing are largely in the downrange direction, with the largest miss at about 0.4 m. The propellant consumption is in a near normal distribution, as seen in Fig. 17. Table 2 provides more detailed statistics on the touchdown conditions. All 3000 trajectories may be considered to have landed successfully at the landing site. The mean propellant consumption reported in Table 2 represents the benchmark for the best propellant performance this mission can ever achieve - the data in Ref. [24] for the same Monte Carlo runs under a different powered descent guidance method show more than 400 kg in mean propellant consumption. It should be stressed that 6DOF simulations will almost certainly show higher misses and consume more propellant.

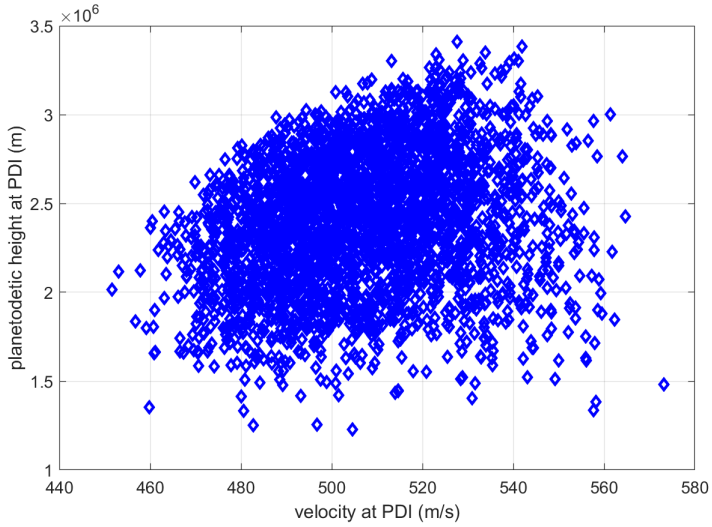


Fig. 15 Dispersions in velocity and altitude at PDI

Table 2 Statistics of touchdown conditions for 3000 Monte Carlo end-to-end simulations

	mean	standard deviation	maximum	minimum
miss (m)	1.66E-02	7.77E-02	4.34E-01	7.91E-03
altitude (m)	6.72E-02	3.63E-02	1.790E-01	-4.74E-02
descending rate (m/s)	2.51E+00	2.90E-02	2.76E+00	2.50E+00
propellant usage (kg)	9.12E+03	2.91E+02	9.98E+03	8.21E+03
powered descent flight time (s)	4.26E+01	1.27E+00	4.63E+01	3.87E+01

XI. Conclusions

The work reported in this paper represents a significant advance in propellant-optimal powered descent guidance algorithm development based on the indirect method. The algorithm presented herein, G-POLAR (“Guidance for Propellant-Optimal Landing of A Rocket"), now determines in a fully automated fashion and without approximations the complete optimal solution, including the optimal thrust switching times, by following strictly the optimal control theory. By a unifying analysis and novel implementation, the algorithm also allows rigorous enforcement of the thrust pointing direction constraint without any increase in algorithmic complexity. On a more intrinsic level, this paper reveals and calls attention to a hidden phenomenon of “solution degradation" in solving the propellant-optimal powered descent problem that can pose a critical safety risk for guidance purposes (as opposed to just offline applications). In

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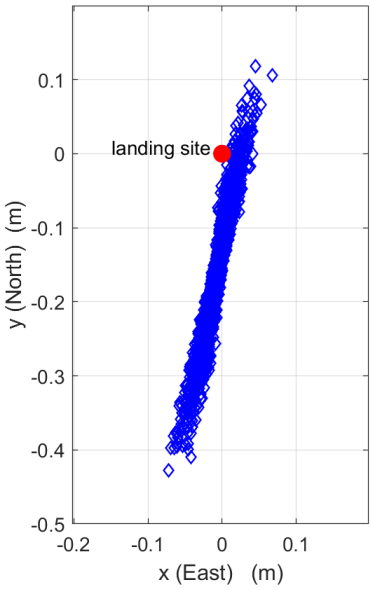


Fig. 16 Touchdown locations of 3000 Monte Carlo trajectories

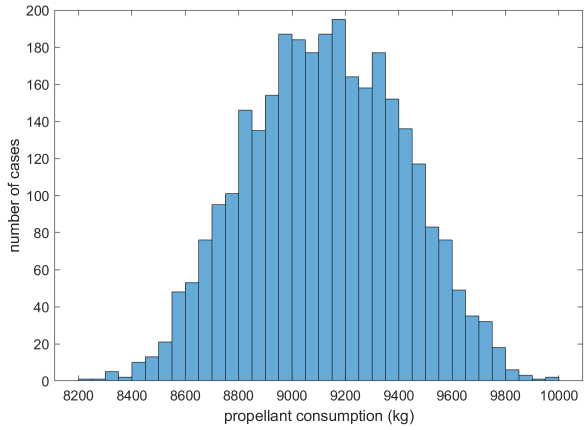


Fig. 17 Propellant consumption distribution

this phenomenon the highly constrained nature of the fuel-optimal powered descent trajectory could result in unintended and unacceptable maneuvers of the vehicle. An effective safeguard utilizing a globally convergent tracking law is developed to mitigate the risk. Independent verification and extensive Monte Carlo simulations offer clear evidence of the promising potential of G-POLAR in propellant-optimal powered descent guidance.

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